FIITJEE Students From All Programs have bagged 34 in Top 100, 66 in Top 200 and 174 in Top 500 All India Ranks. FIITJEE Performance in JEE (Advanced), 2014: 2521 FIITJEE Students from Classroom / Integrated School Programs & 3579 FIITJEE Students from All Programs have qualified in JEE (Advanced), 2014.

ALL INDIA INTEGRATED TEST SERIES

FIITJEE

JEE(Advanced), 2015

ANSWERS, HINTS & SOLUTIONS

HALF COURSE TEST –VIII
(Paper-1)

Q. No.	PHYSICS	CHEMISTRY	MATHEMATICCS
1.	B	B	C
2.	D	C	D
3.	D	A	D
4.	C	D	B
5.	C	A	D
6.	D	C	A
7.	C	C	C
8.	A	D	C
9.	A, B	A, B, C, D	B, D
10.	A, B, D	B, D	A, B, C, D
11.	A, B, C, D	B, C, D	A, B
12.	B, C	B, C, D	A, B
13.	C	A	D
14.	A	B	C
15.	C	D	C
16.	C	B	B
17.	C	C	B
18.	A	A	C
1.	(A) → (q, s) (B) → (p, r) (C) → (p, r) (D) → (q, s)	(A) → (r) (B) → (p, q, r, s,) (C) → (p, s) (D) → (p, q, s)	(A) → (s) (B) → (r) (C) → (q) (D) → (t)
2.	(A) → (r, t) (B) → (t) (C) → (q) (D) → (q)	(A) → (q) (B) → (s) (C) → (p) (D) → (p)	(A) → (p) (B) → (p, q) (C) → (t) (D) → (r)

Physics

PART – I

SECTION – A

1. In one oscillation distance covered = $4 \times 3 = 12$

Here $\omega = \frac{\pi}{2} = \frac{2\pi}{T} \Rightarrow T = 4$ sec

$\therefore$ 8.5 sec = $4 \times 2 + 0.5$ and distance covered = $2 \times 12 + 3 - 3\cos[(\pi/2) \times 0.5] = 27 - \frac{3}{\sqrt{2}}$

3. The line of force due to pressure every where passes through O.

$\therefore$ Net torque = 0

$\Rightarrow$ F = 0

4. Each block completes half the oscillation and time period is independent of amplitude

5. $F = -\frac{dU}{dx} = -U_0\alpha \sin\alpha x = -U_0\alpha \sin\alpha x$

for small angle

$F = -U_0\alpha^2 x$

$T = 2\pi\sqrt{\frac{m}{U_0\alpha^2}}$

6. Since $\frac{m_1}{m_2} = \frac{2}{0.25} = 8 = \frac{\frac{4}{3}\pi r_1^3\rho}{\frac{4}{3}\pi r_2^3\rho}$

$\Rightarrow \frac{r_1}{r_2} = 2$

$v_T \propto r^2$

hence $v_T = \frac{v}{4}$

7. Conservation of linear momentum during the explosion

$mv = \frac{m}{2} \times 0 + \frac{m}{2} \times v' \Rightarrow v' = 2v$

Increase in the mechanical energy = $\Delta K + \Delta U$

$\Delta U = 0 \quad \Delta K = \frac{1}{2}\frac{m}{2}(2v)^2 - \frac{1}{2}mv^2 = \frac{1}{2}mv^2$

$v = \sqrt{\frac{GM}{2R}}$

$\Delta K = \frac{mgR}{4}$

8. Acceleration for particle = g
Acceleration for earth = g/2

$$g_{rel} = g + \frac{g}{2} = \frac{3g}{2}$$

$$h = \frac{1}{2} g_{rel} t^2 \Rightarrow t = \sqrt{\frac{4h}{3g}}$$

9. The satellite will fly out tangentially with same speed.

10. Use vector sum for superposition principle

11. $f_{AC} > f_{AB}$

$\Rightarrow \quad |f_A - f_C| > |f_A - f_B|$

$\Rightarrow$ All of them are possible.

12. Energy stored per unit volume = $\frac{1}{2} \times$ stress $\times$ strain = $\frac{1}{2} \times \frac{F}{A} \times \frac{\Delta \ell}{\ell}$

Total Energy = $\frac{1}{2} \times F \times \Delta \ell = \frac{1}{2} Mg \ell$

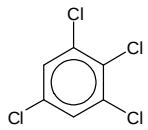
Chemistry

PART – II

SECTION – A

1. The above reaction is not at equilibrium as,
$Q_c > K_c$, So the reaction will move in the backward direction.
Hence **(B)** is correct.

2. Let the dipole moment of chloro benzene = $\mu_1 D_1$ then $x = 2 \times \mu_1 \times \cos 120°/2$
$\therefore \mu_1 = x$ D
Hence **(C)** is correct.



3 nth excited state means that electron is in the $(n + 1)^{th}$ level six, these 4 – H – atoms show a total of six different spectral lines, while returning to the ground level, that is possible only when $(n + 1) = 4$.
Hence **(A)** is correct.

4. Let moles of FeC_2O_4 be x & moles of $FeSO_4$ be y.
Then, $3x + y = 0.05$ ...(i)
also, $x + y = 0.025$ …(ii)
Hence **(D)** is correct.

5. $Sn(50) = 1s^2 2s^2.2p^6 3s^2 3p^6.3d^{10}.4s^2.4p^6.4d^{10}.5s^2.5p^2$ (by obeying Auf – bau rule)
$Sn(50) = 1s^2 2s^2.2p^6 3s^2 3p^6.3d^{10}.4s^2.4p^6.4d^{10}.4f^4$ (if not following Auf – bau rule)
So, if Auf – bau rule is not followed, then Sn would be placed of in f – block.
Hence **(A)** is correct.

6.

$$H_2CO_3 + NaOH \rightleftharpoons NaHCO_3 + H_2O$$

m. mol	10	15	0	0
after the reaction	0	5	10	

Also,

$$NaHCO_3 + NaOH \rightleftharpoons Na_2CO_3 + H_2O$$

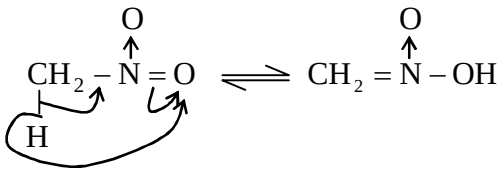
10	5	0
5	0	5

So, it is a buffer,

$$pH = p^{k_{a_2}} + \log \frac{(Na_aCO_3)}{(NaHCO_3)}$$

$\therefore$ pH = $p^{k_{a_2}} = 13$
$\therefore pOH = 1$
Hence **(C)** is correct.

7. The more is the derivation of a real gas from ideal behaviour the more is the critical temperature.
Hence **(C)** is correct.

8. 

10. In case of physical equilibria, equilibria always shifts from low density to high density on increasing the pressure.

11. Since 5.6 littre are occupied by ¼ mole of NTP, so the molecular weight of metal chloride = 133.5 g.
Let the formula of the chloride be MCl_x
$x \times$ mole of metal = $1 \times$ mole of Cl
Let the total mass of metal chloride = 100 g
Then mass of Cl = 80 g
& mass of metal = 20 g
& let mol weight of metal = M_A
Now,

$$x \times \frac{20}{M_A} = 1 \times \frac{80}{35.5} \qquad ...(i)$$

Also, $M_A + 35.5x = 133.5$ ….(ii)
On solving, $x = 3$ & $M_A = 27$ g
Hence (B, C, D)

16. $AgCl(s) \rightleftharpoons Ag^+(aq) + Cl^-(aq)$
$K_{sp} = x^2$
$\therefore x = \sqrt{K_{sp}} = 10^{-5}M$
Hence (B) is correct.

17. $K_{sp} = [Ag^+][Cl]$
$10^{-10} = \left[Ag^+\right]\left[10^{-3}\right] \therefore \left[Ag^+\right] = 10^{-7}M$
Now,

$$\left[Ag(NH_3)_2\right]^+ \rightleftharpoons Ag^+ + 2NH_3$$

Initial conc.	4×10^{-3}	10^{-7}	2×10^{-2}
at equilibrium	$\left(4\times10^{-3}-x\right)$	$\left(10^{-7}+x\right)$	$\left(2\times10^{-2}+2x\right)$
	$\approx 4\times10^{-3}$	$\approx 10^{-7}$	$\approx 2\times10^{-2}$

So, $K_d = \dfrac{\left[Ag^+\right][NH_3]^2}{\left[Ag(NH_3)_2\right]^+} = \dfrac{10^{-7}\times\left(2\times10^{-2}\right)^2}{\left(4\times10^{-3}\right)} = 10^{-8}$
Hence (C) is correct.

Mathematics

PART – III

SECTION – A

1. Let α, β be the roots of $ax^2 - bx - c = 0$ and let α', β' be the roots of $a'x^2 - b'x - c' = 0$ such that $|\alpha - \beta| = |\alpha' - \beta'|$, i.e., $(\alpha + \beta)^2 - 4\alpha\beta = (\alpha' + \beta')^2 - 4\alpha'\beta'$

$\Rightarrow \dfrac{b^2 + 4ac}{a^2} = \dfrac{b'^2 + 4a'c'}{a'^2}$

Hence, the expression $\dfrac{b^2 + 4ac}{a^2}$ does not vary in value

2. Let $z = x + iy$, then $\arg(z - 3i) = \arg(x + iy - 3i) = \dfrac{3\pi}{4}$

$\Rightarrow x < 0, y - 3 > 0 \left(\because \dfrac{3\pi}{4} \text{ is in II quadrant}\right)$

and $\dfrac{y-3}{x} = \tan\dfrac{3\pi}{4} = -1$

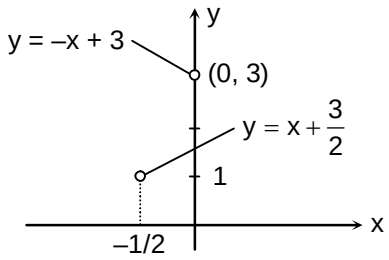
$\Rightarrow y = -x + 3 \ \forall\ x < 0$ and $y > 3$ ..... (1)

and $\arg(2z + 1 - 2i) = \arg\left[(2x + 1) + i(2y - 2)\right] = \dfrac{\pi}{4}$

$\Rightarrow 2x + 1 > 0, 2y - 2 > 0 \left(\because \dfrac{\pi}{4} \text{ is in I quadrant}\right)$

and $\dfrac{2y-2}{2x+1} = \tan\dfrac{\pi}{4} = 1 \Rightarrow 2y - 2 = 2x + 1$

$\Rightarrow y = x + \dfrac{3}{2} \ \forall\ x > -\dfrac{1}{2}, y > 1$ ..... (2)



From equations (1) and (2), we get graph. It is clear from the graph that two lines do not intersect
$\therefore$ No point of intersection

3. Let $a = b - d$ ad $c = b + d$, then $a + b + c = \dfrac{3}{2} \Rightarrow b = \dfrac{1}{2}$

$\therefore$ The numbers are $\dfrac{1}{2} - d$, $\dfrac{1}{2}$, $\dfrac{1}{2} + d$ [$d > 0$ as $a < b < c$]

Now a^2, b^2, c^2 are in G.P. $\Rightarrow \left(b^2\right)^2 - a^2c^2 \Rightarrow \left(\dfrac{1}{2}\right)^2 = \left(\dfrac{1}{2} - d\right)^2\left(\dfrac{1}{2} + d\right)^2$

$\Rightarrow \dfrac{1}{16} = \left(\dfrac{1}{4} - d^2\right)^2 \Rightarrow \dfrac{1}{4} - d^2 = \pm\dfrac{1}{4} \Rightarrow d = \pm\dfrac{1}{\sqrt{2}} \Rightarrow d = \dfrac{1}{\sqrt{2}}$ ($\because d > 0$)

4. $\dfrac{(1 + 24)^{100}}{8} = \dfrac{1}{8} + \text{Integer}$

5. Maximum number of point of intersection = $2 \times {}^8C_2 = 56$

6. $a \sin x + b\{\cos(x + \theta) + \cos(x - \theta)\} = d$
$\Rightarrow a \sin x + 2b \cos x \cos\theta = d$

$$\cos^2\theta \geq \frac{d^2 - a^2}{4b^2} \Rightarrow |\cos\theta| \geq \frac{\sqrt{d^2 - a^2}}{2|b|}$$

7. $\tan^{-1}\sqrt{x^2 - 3x + 2} < \frac{\pi}{2}$ and $\cos^{-1}\sqrt{4x - x^2 - 3} \leq \frac{\pi}{2}$

$\therefore$ 0 < LHS < $\pi \Rightarrow$ no solution

8. $$\left[\left(\frac{a}{2} - c\right)^2 + \left(b - \frac{\sqrt{3}a}{2}\right)^2\right] = 0$$

$$\Rightarrow a = \frac{c}{\left(\frac{1}{2}\right)} = \frac{b}{\left(\frac{\sqrt{3}}{2}\right)}$$

9. $b^2 - 4ac < 0$

$\therefore f(x) = ax^2 + bx + c$ must have same sign for all real x

$f(-1) = a - b + c < 0$

$\therefore f(-2) < 0 \Rightarrow 4a - 2b + c \Rightarrow 4a + c < 2b$

$f\left(-\frac{1}{2}\right) < 0 \Rightarrow \frac{a}{4} - \frac{b}{2} + c < 0 \Rightarrow a - 2b + 4c < 0 \Rightarrow a + 4c < 2b$

10. Circle $(\alpha + r)^2 + y^2 = r^2$

Solving with the parabola, x = 0 and $12 - 2r \neq 0$

$\therefore r_{max} = 6$

11. $$\left|\frac{z_1 + i\frac{z_1 + z_2}{2}}{z_1 - i\left(\frac{z_1 + z_2}{2}\right)}\right| = 1$$

$$\Rightarrow \left|\frac{\frac{2z_1}{z_1 + z_2} + i}{\frac{2z_1}{z_1 + z_2} - i}\right| = 1$$

$\Rightarrow \frac{2z_1}{z_1 + z_2}$ is purely real

$\Rightarrow \frac{z_1 + z_2}{z_1}$ is purely real

$\Rightarrow \frac{z_2}{z_1}$ is purely real

12. Equation is equivalent to $|S_1P - S_2P|$ = constant

$\therefore$ 2a = k and $2ae = S_1S_2 = 2$

$\therefore e = \frac{2}{k}$

Also, k > 0

$\therefore k \in (0, 2)$

13.-15. $\vec{r} = xi + yj$

$x^2 + y^2 + 8x - 10y + 40 = 0$

$c = (-4, 5), r = 1$

$P_1 = \max\{(x + 2)^2 + (y - 3)^2\}$

$P_2 = \min\{(x + 2)^2 + (y - 3)^2\}$

$\therefore P = (-2, 3)$

$CP = 2\sqrt{2}$

$P_2 = \left(2\sqrt{2} - 1\right)^2$, $P_1 = \left(2\sqrt{2} + 1\right)^2$

$\overrightarrow{OA} = 2i + 2j$

$\overrightarrow{OB} = 3i$

$\therefore \overrightarrow{OA} \cdot \overrightarrow{OB} = 3$

16.-18. $\left(z_1 + z_2\right)\left(\overline{z_1 + z_2}\right) = \left|z_1\right|^2 + \left|z_2\right|^2$

$\Rightarrow z_1\bar{z}_2 + \bar{z}_1 z_2 = 0$

$\Rightarrow \frac{z_1}{z_2} + \left(\frac{\bar{z}_1}{\bar{z}_2}\right) = 0$

SECTION – B

1. (A) $\frac{hx}{a^2} + \frac{ky}{b^2} = \frac{h^2}{a^2} + \frac{k^2}{b^2}$ which passes through (0, b)

$\therefore \frac{k}{b} = \frac{h^2}{a^2} + \frac{k^2}{b^2}$

$\Rightarrow$ Ellipse

(B) $h = ae$, $k = a(1 - e^2)$

$\therefore h^2 = -a(k - a)$

$\therefore$ Parabola

(C) $y - mx = \sqrt{a^2m^2 + b^2}$

Perpendicular from (ae, 0) is $my + x = ae$

$\therefore x^2 + y^2 = a^2$

$\therefore$ Circle

(D) $\frac{x}{a} + \frac{y}{b} = 1$ passes through (α, β)

$\therefore \frac{\alpha}{a} + \frac{\beta}{b} = 1$

$h = \frac{a}{2}$, $k = \frac{b}{2}$

$\therefore \frac{\alpha}{h} + \frac{\beta}{k} = 2$

$\frac{\alpha}{x} + \frac{\beta}{y} = 2$

$\Rightarrow \left(x - \frac{\alpha}{2}\right)\left(y - \frac{\beta}{2}\right) = \frac{\alpha\beta}{4}$

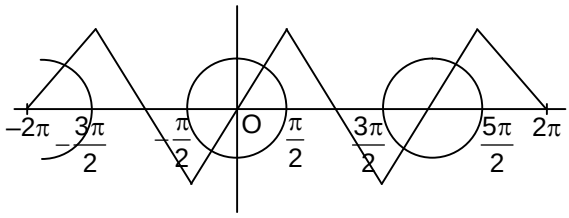
$\therefore$ Rectangular hyperbola

2. (A) $(x-1)^2 + 3\{1 + \sin(ax + b)\} = 0$
$\Rightarrow \alpha = 1$ and $\sin(a + b) = -1$

(B) $\frac{\pi}{2} - \sin^{-1} x \geq \sin^{-1} x$

$\therefore -1 \leq x \leq \frac{1}{\sqrt{2}}$

(C) The graph of $|y| = \cos x$ and $y = \sin^{-1}(\sin x)$ intersect at 5 points in $[-2\pi, 3\pi]$



(D) Adding we get, $\left(\sin^{-1} y\right)^2 = \frac{4n+1}{32}\pi^2$

$\Rightarrow 0 \leq \frac{4n+1}{32}\pi^2 \leq \frac{\pi^2}{4}$

Also, $\cos^{-1} x = \frac{4n-1}{32}\pi^2$

$\therefore n = 1$